Design, Simulation, and 3D Printing of a LOX/RP-1 Bell Nozzle

Independent Engineering Project Report

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Abstract

A bell nozzle was designed for a liquid oxygen and RP-1 rocket engine with a chamber pressure of 100 bar, expanding to 0.7 bar, the approximate atmospheric pressure at an altitude of 3 km. NASA's Chemical Equilibrium with Applications program (CEA) provided the combustion gas properties, an ideal area ratio of 15.92, and a predicted exit Mach number of 3.54. The nozzle contour was modeled in OpenSCAD and printed in PLA on a desktop FDM printer. A steady compressible flow simulation was set up in SimFlow with the OpenFOAM solver, but none of the runs converged. The furthest run reached iteration 24 before diverging, and no usable flow field was produced. The printed geometry matches the CEA area ratio. The simulation is reported as a record of the setup and its failure, along with the likely causes and the changes planned for the next attempt.

1. Objective

The project began as research into why the nozzles on different launch vehicles have different shapes. It had three goals: size a bell nozzle for a specific engine condition using real combustion data, test the design with computational fluid dynamics (CFD), and build a physical model of the final geometry. The work was self-directed.

2. Design Conditions and Thermochemistry

The design point is a LOX/RP-1 engine at 100 bar expanding to 0.7 bar. The exit pressure was matched to the atmosphere at about 3 km so that the nozzle would be ideally expanded at that altitude. Gas properties were calculated with NASA CEA (CEA2, run on 8 October 2025) for equilibrium composition during expansion from an infinite area combustor. The full CEA output is reproduced in Appendix A.

Table 1. *Design inputs.*

Parameter	Value
Propellants	Liquid oxygen (90.19 K) and RP-1 (298.15 K)
Oxidizer to fuel ratio (by mass)	2.30
Chamber pressure	100 bar
Exit pressure	0.70 bar (approximately 3 km altitude)

Parameter	Value
Chamber to exit pressure ratio	142.86
Chemistry model	Equilibrium, infinite area combustor

Table 2. *NASA CEA results at the chamber, throat, and exit.*

Property	Chamber	Throat	Exit
Pressure (bar)	100.00	57.38	0.70
Temperature (K)	3615.6	3397.4	1750.6
Molecular weight (g/mol)	22.40	22.65	23.35
Ratio of specific heats, γ	1.152	1.153	1.219
Speed of sound (m/s)	1243.2	1198.9	871.8
Mach number	0	1.000	3.541

CEA gives an exit to throat area ratio of 15.916, a characteristic velocity of 1812.6 m/s, and an exit velocity of 3086.6 m/s, equal to a specific impulse of 314.7 s at the 0.7 bar design pressure. The vacuum specific impulse is 3288.6 m/s, or 335.3 s.

2.1 Check with the isentropic area relation

As an independent check, the isentropic relation between area ratio and Mach number was solved with a constant ratio of specific heats:

$$A/A^* = (1/M)\,[\,(2/(\gamma+1))\,(1+((\gamma-1)/2)\,M^2)\,]^{(\gamma+1)/(2(\gamma-1))}$$

With γ = 1.2, an area ratio of 15.92 corresponds to Mach 3.60. With the throat value of γ = 1.153, the same area ratio gives Mach 3.43. The CEA value of Mach 3.54 falls between the two because γ rises from 1.15 to 1.22 as the gas cools and its composition changes during expansion, which a single constant γ cannot capture. The original design target of roughly Mach 3.5 is therefore met at the CEA design point.

3. Nozzle Geometry

The nozzle was modeled in OpenSCAD from a profile of six points: the inlet, the throat, three points along the diverging wall, and the exit. The profile was revolved about the centerline with rotate_extrude() and offset by a constant 2.0 mm wall to form a shell. The wall thickness and the overall scale are variables in the script, so the model can be resized without editing the points.

The throat and exit radii were set to give the CEA area ratio. The three intermediate wall points were placed by hand to approximate a bell shape. They were not calculated with the method of

characteristics or a Rao parabolic approximation, so the contour only approximates an ideal bell nozzle. Table 3 lists the inner profile at the 0.81 scale used for printing.

Table 3. *Inner profile points at the printed scale of 0.81 (mm).*

Point	Axial distance from inlet	Inner radius
Inlet	0.00	20.25
Throat	12.15	8.10
Wall point 1	29.97	17.42
Wall point 2	47.79	23.90
Wall point 3	65.61	28.67
Exit	84.48	32.32

The exit to throat radius ratio is 32.32 / 8.10 = 3.99, which gives a geometric area ratio of 15.92, matching the CEA value of 15.916. The printed part is 84.48 mm tall with a 2.0 mm wall.

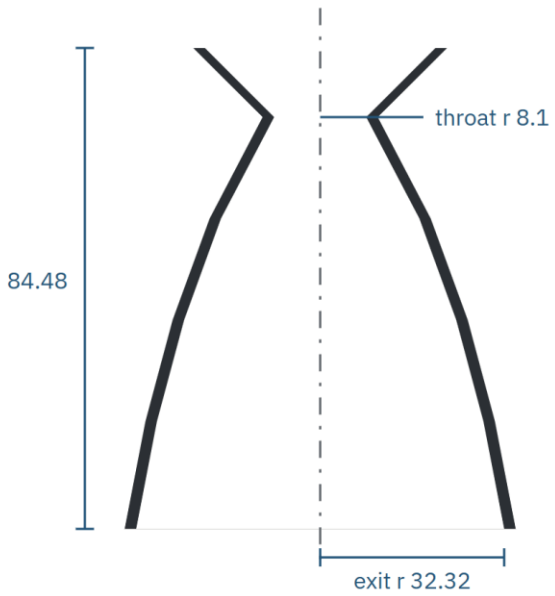


Figure 1. *Cross-section of the nozzle drawn from the six OpenSCAD profile points at the printed scale. Dimensions in millimeters.*

4. CFD Setup

A second OpenSCAD model containing only the internal flow volume, without walls, was exported as the simulation geometry. The case was built in SimFlow 5.0.68, a graphical interface for OpenFOAM v2212, using rhoSimpleFoam, OpenFOAM's steady-state, pressure-based solver

for compressible flow. Results were viewed in ParaView. The free version of SimFlow limits a mesh to 200,000 nodes on a single processor core, and that limit shaped the mesh.

Table 4. *CFD case settings.*

Setting	Value
Software	SimFlow 5.0.68 with OpenFOAM v2212
Solver	rhoSimpleFoam (steady, compressible)
Mesh	Hex box, 40 × 40 × 60, about 96,000 cells, uniform grading
Domain	70 × 70 × 95 mm around the flow volume
Flow model	Laminar
Inlet	Total pressure 100 bar, temperature 3615 K
Outlet	Static pressure 0.7 bar
Walls	No-slip
Discretization	Upwind
Stabilization (later runs)	Temperature limiter

The mesh was the weakest part of the setup. A uniform box places cells of the same size at the throat, where the flow changes fastest, and in the corners of the domain, where it does not change at all. No refinement was applied at the throat or along the wall.

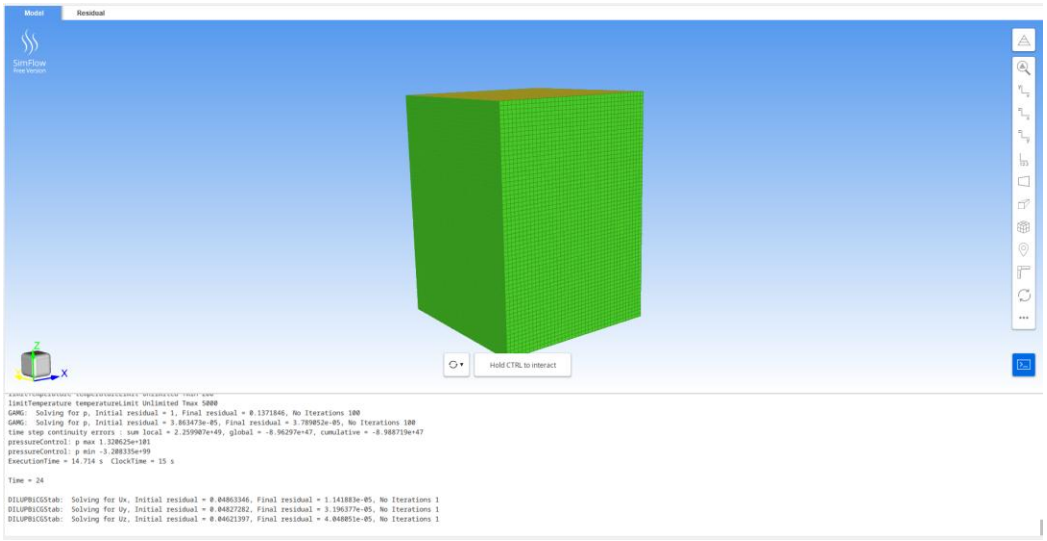


Figure 2. *The 40 × 40 × 60 uniform hex mesh in SimFlow, with the solver log below it.*

5. CFD Results

The runs in Table 5 are the ones recorded on 2 and 3 April 2026. Many more were attempted over the same evening and the following afternoon, each after a change to the mesh, the pressure and temperature settings, or the boundary conditions.

Table 5. *Recorded run history, April 2026.*

Time	Result
2 April, 22:23	Aborted at iteration 2, fatal error in the temperature solve
2 April, 22:28	Same abort, with 1000 iterations requested
2 April, 23:03	Iteration 1, laminar case, no limiters
2 April, 23:19	Iteration 2, continuity error already growing
2 April, 23:29	Iteration 20, with the temperature limiter in place
3 April, 16:49	Iteration 24, the furthest any run reached, then diverged

The early runs aborted inside the thermodynamics within one or two iterations, reporting a temperature of about -1×10^{17} K. A negative temperature of that size means the energy equation had produced a meaningless result, and the solver stopped. Adding a temperature limiter kept the case running to iteration 20 and later to iteration 24, although it suppressed the symptom without removing its cause.

In the final run, the solver reported a continuity error on the order of 10^{49}, a maximum pressure of 1.32×10^{101}, and a minimum pressure of -3.21×10^{99}. In the residual plot (Figure 3), the three velocity residuals fall at first and then wander between about 0.01 and 0.05, while the enthalpy

residual stays at 1 for the whole run. A converging case would show all of the residuals falling together.

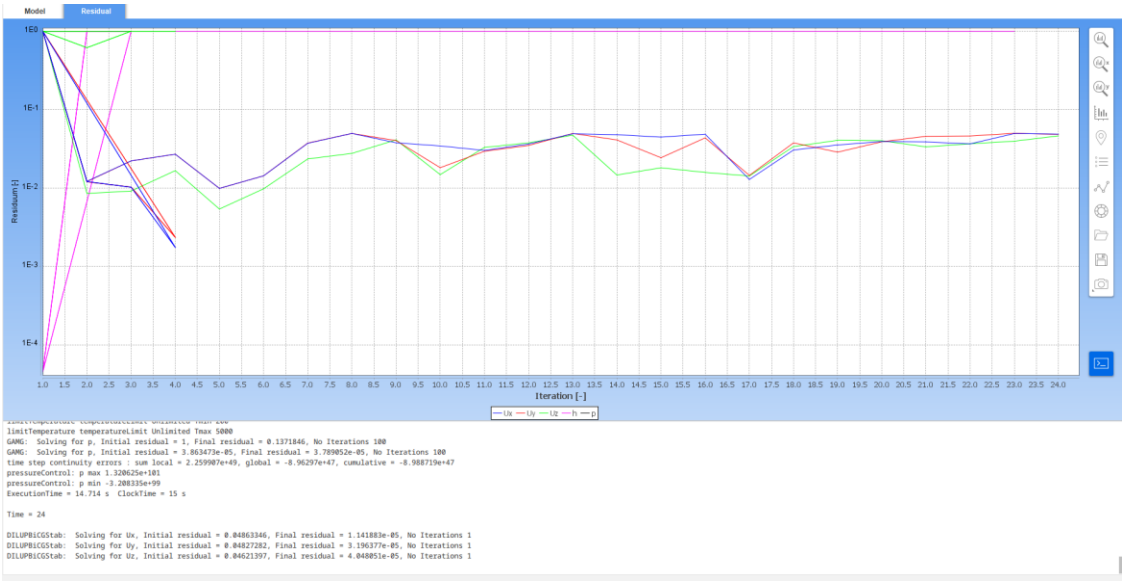


Figure 3. *Residuals for the run on 3 April, which reached iteration 24. The solver log below the plot shows the diverged pressure values.*

No velocity, Mach number, or pressure field was written to disk. Opening the case in ParaView shows only time step zero, which is the mesh before any solution. As a result, choking at the throat, supersonic flow in the bell, and whether the nozzle is over or under expanded could not be confirmed from the simulation.

6. Discussion

The exact cause of the divergence was not identified. The most likely contributors, none of which were tested individually, are the following:

- The uniform mesh is far too coarse at the throat, where the flow passes Mach 1 and the gradients are largest.
- The case started with a 100 bar inlet against ambient initial conditions, with no gradual ramp in pressure.
- The inlet, outlet, and wall patches were assumed, but not verified, to be separate surfaces with the correct condition on each.
- A steady-state solver was asked to settle a flow that first has to establish itself through a strong start-up transient.

The only quantitative result that can be supported is the geometric comparison in Section 3: the printed nozzle has an area ratio of 15.92, against the CEA value of 15.916. None of the CFD output can be used to judge the design.

7. Fabrication

The nozzle was sliced in Cura and printed in PLA on an Entina TINA2S FDM printer. The main decision was print orientation. Printed with the inlet down, the inside of the bell becomes an overhang, and the slicer fills the expansion section with support material that leaves marks on the inner wall, the surface the gas would flow along. Printed upside down, with the exit face on the build plate, each section of wall stays within its self-supporting angle, so no support is needed inside the bell. A raft was still used under the part where it meets the bed.

Table 6. *Print settings.*

Setting	Value
Printer	Entina TINA2S (FDM)
Material	PLA
Layer height	0.2 mm, 422 layers
Infill	10%
Print time	4 h 42 min, 11.2 m of filament
Orientation	Inverted, exit face on the build plate, with a raft



Figure 4. *The printed nozzle standing on its exit face, 84.48 mm tall. Layer lines from the FDM process are visible on the surface.*

8. Conclusions

The printed geometry matches the CEA design point, which gives an exit Mach number of 3.54 at an area ratio of 15.92, and printing the part inverted removed the need for support material inside

the bell. The simulation fell short of its goal. No run converged past iteration 24, and without a flow field the CFD says nothing about how the nozzle would perform.

The most useful outcome of the CFD work was learning how sensitive a compressible simulation is to the mesh, the boundary conditions, and the way the case is started. Small changes in those settings decided whether a run lasted one iteration or twenty-four.

9. Recommendations for Future Work

1. Build a mesh fitted to the nozzle, with small cells at the throat and along the wall and larger cells elsewhere, which also keeps the case under the 200,000 node limit.
2. Check that the inlet, outlet, and wall are separate patches with the correct boundary condition on each.
3. Ramp the inlet pressure up gradually instead of starting at 100 bar.
4. Run a transient case first so the flow can establish before a steady-state solution is attempted.
5. Treat the nozzle as simulated only once the solver produces a Mach number field that can be inspected.

References

[1] Gordon, S., and McBride, B. J. Computer Program for Calculation of Complex Chemical Equilibrium Compositions and Applications, Part I: Analysis. NASA Reference Publication 1311, 1994.

[2] McBride, B. J., and Gordon, S. Computer Program for Calculation of Complex Chemical Equilibrium Compositions and Applications, Part II: Users Manual and Program Description. NASA Reference Publication 1311, 1996.

[3] NASA Glenn Research Center. Chemical Equilibrium with Applications (CEA2). Online run, 8 October 2025.

[4] OpenCFD Ltd. OpenFOAM, version v2212.

[5] SimFlow Technologies. SimFlow, version 5.0.68.

Appendix A. NASA CEA Output

Input summary and equilibrium rocket performance from the CEA run used for the design (8 October 2025). Mass fractions of the combustion products are omitted.

```
### CEA analysis performed on Wed 08-Oct-2025 19:41:45

 # Problem Type: "Rocket" (Infinite Area Combustor)

 prob case = _______________2254 ro equilibrium

 # Pressure (1 value):
 p,bar= 100
 # Chamber/Exit Pressure Ratio (1 value):
 pi/p= 142.86

 # Oxidizer/Fuel Wt. ratio (1 value):
 o/f = 2.3

 # You selected the following fuels and oxidizers:
 reac
 fuel RP-1              wt%=100.0000  t,k= 298.150
 oxid O2(L)             wt%=100.0000  t,k=  90.190

 # You selected these options for output:
 # short version of output
 output short
 # Proportions of any products will be expressed as Mass Fractions.
 output massf
 # Heat will be expressed as siunits
 output siunits

 # Input prepared by this script:/var/www/sites/cearun/cgi-bin/CEARUN/prepareInpu
 tFile.cgi

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

 Pin =  1450.4 PSIA
 CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
 FUEL        RP-1                        1.0000000    -24717.700    298.150
 OXIDANT     O2(L)                       1.0000000    -12979.000     90.190

 O/F=    2.30000  %FUEL= 30.303030  R,EQ.RATIO= 1.480724  PHI,EQ.RATIO= 1.480724

                CHAMBER   THROAT     EXIT
 Pinf/P           1.0000   1.7428   142.86
 P, BAR           100.00   57.379  0.69999
 T, K            3615.59  3397.39  1750.60
 RHO, KG/CU M   7.4524 0 4.6014 0 1.1230-1
 H, KJ/KG         -818.62 -1537.36 -5582.32
 U, KJ/KG        -2160.47 -2784.36 -6205.65
 G, KJ/KG        -42084.6 -40312.9 -25562.6
 S, KJ/(KG)(K)    11.4134  11.4134  11.4134

 M, (1/n)          22.403   22.652   23.351
 (dLV/dLP)t      -1.02291 -1.01693 -1.00003
 (dLV/dLT)p        1.3971   1.3125   1.0010
 Cp, KJ/(KG)(K)    4.6835   4.2314   1.9841
 GAMMAs            1.1518   1.1527   1.2192
 SON VEL,M/SEC     1243.2   1198.9    871.8
 MACH NUMBER        0.000    1.000    3.541

 PERFORMANCE PARAMETERS

 Ae/At                     1.0000   15.916
```

CSTAR, M/SEC	1812.6	1812.6
CF	0.6614	1.7028
Ivac, M/SEC	2239.0	3288.6
Isp, M/SEC	1198.9	3086.6